Siddhesh Raut (CS23155) Assignment 4

```
import pandas as pd
from sklearn.metrics import accuracy_score
from sklearn.tree import DecisionTreeClassifier
```

```
from google.colab import drive
drive.mount("/content/drive")
```

```
Mounted at /content/drive
```

```
path="/content/drive/MyDrive/ML_Data/Developer_Performance_dataset_1000 - Developer_Performance_dataset_1000.cs
dF=pd.read_csv(path)
df.head(5)
```

	Hours_Coding	Lines_of_Code	Bugs_Found	Bugs_Fixed	AI_Usage_Hours	Sleep_Hours	Cognitive_Load	Coffee_Intake	Stress_Level
0	7	416	9	7	6	5.9	92	7	99
1	4	269	16	13	5	5.1	85	2	100
2	11	439	3	0	2	6.2	38	2	55
3	8	472	18	9	4	4.2	26	5	30
4	5	265	19	16	5	8.1	82	6	82

```
df.info()
```

Show hidden output

```
df.shape
```

```
(1900, 13)
```

```
df.describe()
```

Show hidden output

```
df.columns
```

Show hidden output

```
df.isnull().any().any()
```

```
np.False_
```

```
df.isnull().sum()
```

Show hidden output

```
#print missing value
df[df.isnull().any(axis=1)]
```

	Hours_Coding	Lines_of_Code	Bugs_Found	Bugs_Fixed	AI_Usage_Hours	Sleep_Hours	Cognitive_Load	Coffee_Intake	Stress_Level

```
df.drop('number', axis=1, inplace=True, errors='ignore')
df.columns
```

```
Index(['Hours_Coding', 'Lines_of_Code', 'Bugs_Found', 'Bugs_Fixed',
       'AI_Usage_Hours', 'Sleep_Hours', 'Cognitive_Load', 'Coffee_Intake',
       'Stress_Level', 'Task_Duration_Hours', 'Commits', 'Errors',
       'Task_Success_Rate'],
      dtype='object')
```

```
df.shape
```

```
(1000, 13)
```

```
before_rows=df.shape[0]
df = df.dropna()
after_rows = df.shape[0]
print("Before rows:",before_rows)
print("After rows:",after_rows)
print("The number of dropped rows are:",before_rows-after_rows)
```

```
Before rows: 1000
After rows: 1000
The number of dropped rows are: 0
```

```
df.shape
```

```
(1000, 13)
```

```
clean_data = df.copy()
# clean_data ['high_humidity_label'] = (clean_data['relative_humidity_3pm']>24.99)*1
# print(clean_data['high_humidity_label'])
```

```
clean_data.columns
```

Show hidden output

```
clean_data.drop('relative_humidity_9am', axis=1, inplace=True, errors='ignore')
clean_data.drop('relative_humidity_3pm', axis=1, inplace=True, errors='ignore')
clean_data.columns
```

```
Index(['Hours_Coding', 'Lines_of_Code', 'Bugs_Found', 'Bugs_Fixed',
       'AI_Usage_Hours', 'Sleep_Hours', 'Cognitive_Load', 'Coffee_Intake',
       'Stress_Level', 'Task_Duration_Hours', 'Commits', 'Errors',
       'Task_Success_Rate'],
      dtype='object')
```

```
X=clean_data.iloc[:,:-1]
print(X)
```

Show hidden output

```
Y=clean_data.iloc[:,-1]
print(Y)
```

Show hidden output

```
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3, random_state=42)
```

```
X_train
```

	Hours_Coding	Lines_of_Code	Bugs_Found	Bugs_Fixed	AI_Usage_Hours	Sleep_Hours	Cognitive_Load	Coffee_Intake	Stress_Leve
541	3	178	0	0	1	8.1	28	0	3
440	6	226	5	2	2	5.2	21	2	3
482	5	388	4	1	6	5.4	68	1	7
422	1	176	8	3	3	6.4	84	3	9
778	5	340	17	15	0	6.3	24	4	4
...	...	...	...	...	...	...	...	...	
106	8	236	17	15	3	6.0	61	4	7
270	8	228	5	3	2	8.8	23	5	3
860	1	102	19	14	2	5.9	51	3	6
435	7	189	17	12	0	8.1	26	0	4
102	10	282	4	0	2	8.4	76	0	8

700 rows × 12 columns

```
X_test
```

	Hours_Coding	Lines_of_Code	Bugs_Found	Bugs_Fixed	AI_Usage_Hours	Sleep_Hours	Cognitive_Load	Coffee_Intake	Stress_Leve
521	3	234	5	3	2	8.4	63	5	8
737	2	254	5	1	6	6.6	86	4	10
740	5	266	13	9	4	6.4	84	3	8
660	5	140	3	0	0	8.1	60	3	7
411	8	810	2	0	6	8.0	43	5	5
...	...	...	...	...	...	...	...	...	
468	4	168	8	6	3	4.4	60	5	7
935	4	210	2	0	2	7.9	34	0	4
428	9	646	8	2	2	8.3	94	7	10
7	3	234	5	1	6	5.8	78	2	8
155	7	224	19	15	1	8.0	54	2	6

300 rows × 12 columns

```
Y_train
```

	Task_Success_Rate
541	85
440	80
482	50
422	30
778	74
...	...
106	43
270	79
860	52
435	84
102	30

700 rows × 1 columns

dtype: int64

```
Y_test
```

	Task_Success_Rate
521	39
737	36
740	36
660	49
411	82
...	...
468	49
935	77
428	30
7	40
155	49

300 rows × 1 columns

dtype: int64

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler = scaler.transform(X_test)
```

```
humidity_classifier = DecisionTreeClassifier(max_leaf_nodes=10, random_state=0)
humidity_classifier.fit(X_train, Y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(max_leaf_nodes=10, random_state=0)
```

```
import matplotlib.pyplot as plt
from sklearn import tree
plt.figure(figsize=(10,10))
tree.plot_tree(humidity_classifier, filled=True)
```

```
[Text(0.5416666666666666, 0.9285714285714286, 'x[6] <= 1.108\ngini = 0.959\nsamples = 700\nvalue = [120, 10, 8, 9, 10, 6, 6, 5,
6, 13, 11, 9, 14\n13, 7, 8, 9, 3, 10, 6, 9, 13, 16, 9, 9, 5\n12, 8, 11, 6, 9, 8, 12, 13, 10, 6, 4, 11, 3\n8, 7, 11, 14, 11, 11,
10, 9, 5, 12, 16, 6, 8\n7, 8, 7, 10, 7, 14, 4, 7, 9, 5, 7, 6, 2, 5\n3, 2, 3, 2, 7]'),
 Text(0.3333333333333333, 0.7857142857142857, 'x[6] <= -0.216\ngini = 0.983\nsamples = 569\nvalue = [20.0, 4.0, 3.0, 5.0, 5.0,
6.0, 3.0, 5.0, 4.0, 11.0\n9.0, 9.0, 12.0, 13.0, 7.0, 8.0, 9.0, 3.0, 10.0, 6.0\n9.0, 13.0, 16.0, 9.0, 9.0, 5.0, 12.0, 8.0, 11.0,
6.0\n9.0, 8.0, 12.0, 13.0, 10.0, 6.0, 4.0, 11.0, 3.0, 8.0\n7.0, 11.0, 14.0, 11.0, 11.0, 10.0, 9.0, 5.0, 12.0\n16.0, 6.0, 8.0,
7.0, 8.0, 7.0, 10.0, 7.0, 14.0, 4.0\n7.0, 9.0, 5.0, 7.0, 6.0, 2.0, 5.0, 3.0, 2.0, 3.0\n2.0, 7.0]'),
 Text(0.4375, 0.8571428571428572, 'True  '),
 Text(0.16666666666666666, 0.6428571428571429, 'x[8] <= -0.868\ngini = 0.974\nsamples = 319\nvalue = [0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0\n0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0\n0.0, 1.0, 5.0, 1.0, 1.0, 1.0, 3.0, 5.0, 7.0,
2.0\n6.0, 3.0, 9.0, 9.0, 9.0, 2.0, 3.0, 9.0, 3.0, 7.0\n7.0, 10.0, 13.0, 11.0, 11.0, 10.0, 9.0, 5.0, 12.0\n16.0, 6.0, 8.0, 7.0,
8.0, 7.0, 10.0, 7.0, 14.0, 4.0\n7.0, 9.0, 5.0, 7.0, 6.0, 2.0, 5.0, 3.0, 2.0, 3.0\n2.0, 7.0]'),
 Text(0.08333333333333333, 0.5, 'gini = 0.961\nsamples = 172\nvalue = [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0\n0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0\n0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0\n0.0, 0.0, 1.0, 0.0, 0.0, 1.0,
0.0, 1.0, 1.0, 3.0\n2.0, 4.0, 5.0, 5.0, 3.0, 6.0, 6.0, 2.0, 8.0, 12.0\n3.0, 3.0, 5.0, 5.0, 5.0, 10.0, 4.0, 14.0, 4.0, 7.0\n9.0,
5.0, 7.0, 6.0, 2.0, 5.0, 3.0, 2.0, 3.0, 2.0\n7.0]'),
 Text(0.25, 0.5, 'gini = 0.962\nsamples = 147\nvalue = [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0\n0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0\n0.0, 1.0, 5.0, 1.0, 1.0, 1.0, 3.0, 5.0, 6.0, 2.0\n6.0, 3.0, 8.0, 9.0, 9.0, 1.0, 3.0, 8.0, 2.0,
4.0\n5.0, 6.0, 8.0, 6.0, 8.0, 4.0, 3.0, 3.0, 4.0, 4.0\n3.0, 5.0, 2.0, 3.0, 2.0, 0.0, 3.0, 0.0, 0.0, 0.0\n0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0\n0.0]'),
 Text(0.5, 0.6428571428571429, 'x[6] <= 0.651\ngini = 0.965\nsamples = 250\nvalue = [20, 4, 3, 5, 5, 6, 3, 5, 4, 11, 9, 9, 12,
13\n7, 8, 9, 3, 10, 6, 9, 12, 11, 8, 8, 4, 9, 3\n4, 4, 3, 5, 3, 4, 1, 4, 1, 2, 0, 1, 0, 1\n1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.4166666666666667, 0.5, 'gini = 0.962\nsamples = 162\nvalue = [1, 0, 0, 2, 0, 0, 2, 3, 0, 7, 5, 6, 6, 9\n4, 4, 7, 2, 8,
4, 7, 11, 10, 6, 8, 4, 9, 3\n4, 4, 3, 5, 3, 4, 1, 4, 1, 2, 0, 1, 0, 1\n1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.5833333333333334, 0.5, 'x[4] <= 0.252\ngini = 0.92\nsamples = 88\nvalue = [19, 4, 3, 3, 5, 6, 1, 2, 4, 4, 4, 3, 6,
4\n3, 4, 2, 1, 2, 2, 2, 1, 1, 2, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.5, 0.35714285714285715, 'x[5] <= -0.368\ngini = 0.834\nsamples = 53\nvalue = [19, 4, 2, 2, 4, 5, 0, 2, 2, 3, 3, 2, 3,
0\n1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.4166666666666667, 0.21428571428571427, 'x[4] <= -0.745\ngini = 0.653\nsamples = 21\nvalue = [11, 0, 0, 0, 1, 5, 0, 1,
0, 0, 0, 0, 2, 0\n1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.3333333333333333, 0.07142857142857142, 'gini = 0.298\nsamples = 11\nvalue = [9, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.5, 0.07142857142857142, 'gini = 0.8\nsamples = 10\nvalue = [2, 0, 0, 0, 1, 3, 0, 1, 0, 0, 0, 0, 2, 0\n1, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.5833333333333334, 0.21428571428571427, 'gini = 0.877\nsamples = 32\nvalue = [8, 4, 2, 2, 3, 0, 0, 1, 2, 3, 3, 2, 1,
0\n0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.6666666666666666, 0.35714285714285715, 'gini = 0.94\nsamples = 35\nvalue = [0, 0, 1, 1, 1, 1, 1, 0, 2, 1, 1, 1, 3,
4\n2, 3, 2, 1, 2, 2, 2, 1, 1, 2, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.75, 0.7857142857142857, 'x[4] <= 0.252\ngini = 0.41\nsamples = 131\nvalue = [100, 6, 5, 4, 5, 0, 3, 0, 2, 2, 2, 0, 2,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.6458333333333333, 0.8571428571428572, '  False'),
 Text(0.6666666666666666, 0.6428571428571429, 'gini = 0.121\nsamples = 79\nvalue = [74, 3, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.8333333333333334, 0.6428571428571429, 'x[5] <= 0.595\ngini = 0.716\nsamples = 52\nvalue = [26, 3, 5, 4, 5, 0, 2, 0, 1,
2, 2, 0, 2, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\n0]'),
 Text(0.75, 0.5, 'gini = 0.593\nsamples = 39\nvalue = [24, 0, 4, 2, 4, 0, 1, 0, 0, 1, 1, 0, 2, 0\n0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```